Evaluating the Robustness of rStar: A Novel Framework for Enhanced Reasoning in Small Language Models

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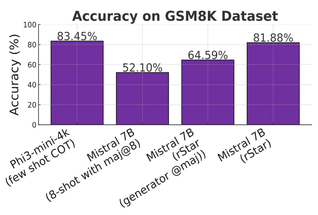
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Introduction

- **Small language models** (SLMs) show strong reasoning abilities, but benchmarks like **Grade School Math 8K** (GSM8K) may overestimate their true reasoning capabilities [1]
- Recent methods **enhance reasoning** but **may neglect robustness** to diverse input variations [2]
- This study evaluates the **robustness of rStar** by evaluating its performance across diverse variations of inputs to identify its **strengths** and **limitations**, offering a more accurate assessment of its reasoning abilities in **mathematical problem-solving**.

GSM8K

Q: John buys purple pairs of shoes for each of his 3 children. They cost 60 each. How much did he pay?
A: "He bought $3 * 2 =<<3*2=6>>6$ pairs of shoes So he spent $60 * 6 = \$ <<60*6=360>>360$ #### 360"

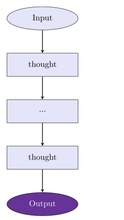


rStar Background

Prompting Language Models to Reason
- Chain of Thought (CoT)

Sampling Reasoning Paths
- Self-consistency

Answer Verification
- **@maj**
- Self verification
- **Mutual consistency**



Chain of Thought-Prompting (CoT)

Related Work

- GSM1k (newly crafter analogous to GSM8k – up to 8% drop in accuracy) [3]
- **Token bias** with logical problems [4]
  - **Token bias** is the idea that the models are biased to the specific tokens that appear in the GSM8k dataset (the names and numbers)
- PMC benchmark – Problems with missing and contradictory conditions [5]

References

[1] Mirzadeh, I., Alizadeh, K., Shahrokhi, H., Tuzel, O., Bengio, S., & Farajtabar, M. (2024). *GSM-Symbolic: Understanding the Limitations of Mathematical Reasoning in Large Language Models*. arXiv. https://arxiv.org/abs/2410.05229

[2] Zhenting Qi and Mingyuan Ma and Jiahang Xu and Li Lyna Zhang and Fan Yang and Mao Yang (2024), Mutual Reasoning Makes Smaller LLMs Stronger Problem-Solvers

[3] H. Zhang, J. Da, D. Lee, V. Robinson, C. Wu, W. Song, T. Zhao, P. Raja, C. Zhuang, D. Slack, Q. Lyu, S. Hendryx, R. Kaplan, M. Lunati, and S. Yue, "A Careful Examination of Large Language Model Performance on Grade School Arithmetic," *arXiv preprint arXiv:2405.00332*, 2024. [Online]. Available: https://arxiv.org/abs/2405.00332

[4] B. Jiang, Y. Xie, Z. Hao, X. Wang, T. Mallick, W. J. Su, C. J. Taylor, and D. Roth, "A Peek into Token Bias: Large Language Models Are Not Yet Genuine Reasoners," in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, Y. Al-Onaizan, M. Bansal, and Y.-N. Chen, Eds., Miami, Florida, USA, Nov. 2024, pp. 4722–4756. Association for Computational Linguistics. [Online]. Available: https://aclanthology.org/2024.emnlp-main.272. doi: 10.18653/v1/2024.emnlp-main.272

[5] S.-Y. Tian, Z. Zhou, L.-H. Jia, L.-Z. Guo, and Y.-F. Li, "Robustness Assessment of Mathematical Reasoning in the Presence of Missing and Contradictory Conditions," *arXiv preprint arXiv:2406.05055*, 2024. [Online]. Available: https://arxiv.org/abs/2406.05055

ChatGPT was used to explain complicated topics and help with text refinement

Methodology

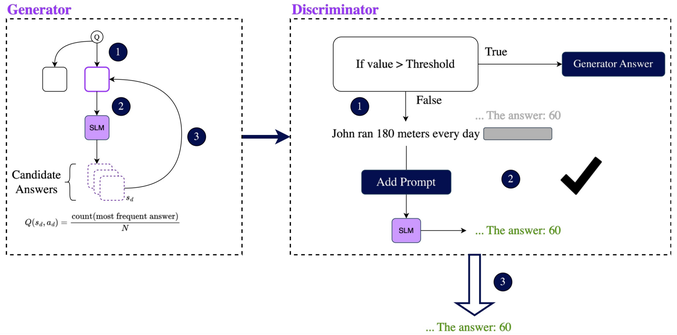
rStar

Generator Phase
1. Selection $UCT(s,a) = \frac{Q(s,a)}{N(s,a)} + c\sqrt{\frac{\ln N_{\text{parent}}(s)}{N(s,a)}}$
2. Expansion & Simulation
3. Propagation $Q(s_i,a) = Q(s_i,a) + Q(s_d,a_d)$

Discriminator Phase
1. Trajectory Masking & Completion
2. Validation
3. Final Trajectory Selection

Actions:
- Propose a one-step thought (A1)
- Propose the remaining thought steps (A2)
- Propose next sub-question along with its answer (A3)
- Answer the sub-question again (A4)

Generator SLM: Mistral 7B-v0.1
Discriminator SLM: Phi-3-mini-4k-instruct



Variations

- Enhance GSM8K dataset by introducing 5 new variation types
- Each variation type is represented by generic templates
- Generate novel math questions derived from the original GSM8K dataset

Template (Names + Numbers)

A company sold {total_gallons} gallons of milk in jars to Mr. {name}'s store at the cost of ${price_per_gallon} per gallon. However, Mr. {name} later realized {exp.frac_num} / {exp.frac_den} of the amount of milk he purchased had passed the expiry date and could not be sold. He returned the sour milk to the company and ordered a refund. Calculate how much he got in refunds.

#Variables:
- name = sample(names)
- total_gallons = range(1000, 5000)
- price_per_gallon = range(2, 5)
- exp.frac_num = range(1, 4)
- exp.frac_den = range(5, 8)

#Conditions:
- expired_gallons = range(1, 1000)
- refund_amount = range(1, 5000)

#Calculations:
expired_fraction = exp.frac_num / exp.frac_den
expired_gallons = expired_fraction × total_gallons
refund_amount = price_per_gallon × expired_gallons

#Solution: If Mr. {name} purchased {total_gallons} gallons, the number of expired gallons is: {exp.frac_num} / {exp.frac_den} × {total_gallons} =

<< {exp.frac_num} / {exp.frac_den} × {total_gallons} = {expired_gallon

{expired_gallons} gallons.
If each gallon costs ${price_per_gallon}, and he returned {expired_gallons} gallons for a refund, the total refund amount is:

<< {price_per_gallon} × {expired_gallons} = {refund_amount} >>

Final Answer: $refund_amount

- **Experiment 1: Only Names**
- **Experiment 2: Only Numbers**
- **Experiment 3: Mixed**

Template (NoOps + Irrelevant Informations)

A company sold 4000 gallons of milk in jars to Mr. Marcellus' store at the cost of $3.5 per gallon. However, Mr. Marcellus later realized 2/5 of the amount of milk he purchased had passed the expiry date and could not be sold. He returned the sour milk to the company and ordered a refund. {sentence} Calculate how much he got in refunds.

- **Experiment 4: NoOps**
{sentence} = {"The milk was delivered in eco-friendly jars.", Mr. Marcellus' store is known for its strict quality control.", ..., }
- **Experiment 5: Irrelevant informations**
{sentence} = {"A marathon is 42.195 kilometers long.", "The Great Wall of China is over 21,000 kilometers long.", ..., }

Conclusion

- **Lower Performance**: All experiments, apart from changing names, resulted in a drop in rStar's performance beyond one standard deviation.
- **Lack of Robustness**: Experiments reveal that rStar is not robust to various input variations.
- **Largest Accuracy Drop**: Numeric variations caused the largest performance decline.
- **Combined Variations**: Combining names and numbers led to a much smaller drop in accuracy compared to numbers alone.
- **NoOps vs. Irrelevant Info**: NoOps caused a slightly larger accuracy drop than irrelevant information, suggesting rStar handles ignoring irrelevant information better than handling seemingly relevant information.

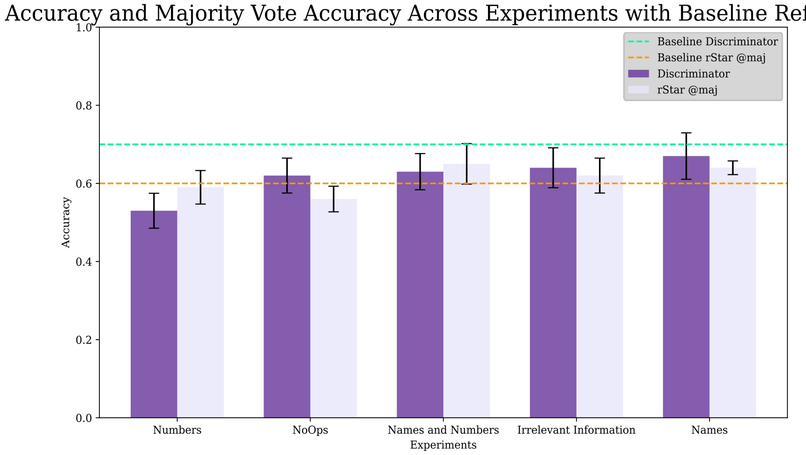
Future Work

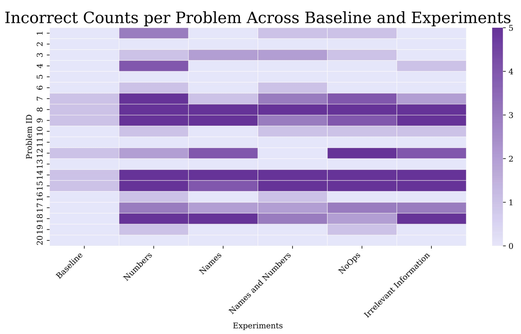
- More variations & bigger subset of questions
- Develop another action that is adapted to the newly introduced variances
- Run experiment more times to account for the inherent variance of language models

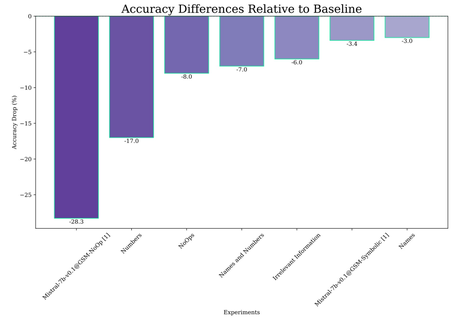
Limitations

- Generator on 40 questions took approx. 12-13 hours using a Tesla A100 PCIE 40 GB
- Discriminator on 40 questions took approx. 6 hours using a Tesla A100 PCIE 40 GB
- Queue time during the busiest periods was approx. 1-2 days

Results







Example of the solution with NoOps

- **Question:**
Michael is traveling on the interstate at an average speed of 50km/hr (taking into account all the necessary stops for gas, relaxation, etc.). If he goes on like this for 3 days, and the distance between Alaska and Texas is 6000 km, what percentage of this distance has he covered? Michael's car is equipped with a GPS system. What percentage of the distance has he covered?",
- **Correct answer: 60%**

- **Question:**
What percentage of the distance has Michael covered with the GPS system? ",
- **Final answer:**
With the GPS system, he knows his exact position. So the distance covered equals the distance between his starting point and his current location .
The answer is 100%.

Example of the solution with irrelevant information

- **Question:**
There are about 7.8 billion people on Earth. Audrey has to take two math tests to pass 6 grade. They must correctly answer 70% of the total questions to move on to the next grade. The first test has 70 questions, and they get 60% of them correct. The second test has 40 questions. How many questions do they need to get right on the second test to pass?
- **Correct answer: 35**

- **Final Answer:**
Let's think step by step. There are about 7.8 billion people on Earth. To pass 6th grade, one must correctly answer 70% of the total questions. So, to pass 6th grade, one needs to answer 70% of 7.8 billion which is 5.46 billion. The first test has 70 questions. One gets 60% of the first test which is 42 questions correct. Therefore, 5.46 billion / 42 is equal to 130 questions correct.
The answer is: 130.